DAY 2 · FRIDAY 23 OCTOBER 2026 · MFB 545

Payments and digital money

A card payment involves five parties in under two seconds; a mobile money withdrawal in Dakar depends on the liquidity of a neighbourhood agent; a CHF 200 transfer costs anything from under 1% to over 10% depending on the channel. This day follows the money step by step: the card circuit and who earns what, the IBAN and the QR-bill, the agent network, four ways to send money to Dakar, a payment API in test mode — then a three-way negotiation, and your fintech's payment flow.

🕘 **09:00–12:15 · 13:30–16:45 · four 90-min blocks** | 💻 **laptop · no programming**

🎤 **the talk is short — 🛠️ most of the time, you do the work**

BY THE END OF DAY 2, YOU WILL BE ABLE TO

- Follow a card payment from authorisation to settlement, and split the merchant discount rate between issuer, scheme and acquirer.
- Check an IBAN, compute the check digit of a QR reference, and read the content of a QR-bill.
- Explain how electronic money is backed, and manage the liquidity of a mobile money agent.
- Compute the total cost of an international transfer — displayed fees *and* exchange-rate margin — across four real channels.
- Send requests to a payment API in test mode and read its responses, errors and webhooks.
- Describe your fintech's payment flow and revenue model.

BEFORE YOU START — TICK

- ☐ **Signed in** on this site (the button at the bottom right).
- ☐ **Day 1's work** at hand: your team's problem and customer (Day 1, section 1.9).

- [ ] **A spreadsheet open** for Block 2. Nothing to install: the API workshop in Block 3 runs on this page.

BLOCK 1 · 09:00–10:30 anatomy of a payment · interchange, IBAN, QR-bill

2.1 **Anatomy of a payment**
Four corners · authorisation, clearing, settlement · who earns what · 3-D Secure · instant payment and QR-bill
🎤 TALK · 20 MIN

2.2 **Interchange · IBAN · QR-bill**
The split simulator · checking an IBAN · the QR reference
🛠 WORKSHOP · 55 MIN

BLOCK 2 · 10:45–12:15 mobile money · Geneva → Dakar · the quiz

2.3 **Mobile money and transfers**
Electronic money · the agent and their two stocks · the true cost of a transfer
🎤 TALK · 15 MIN

2.4 **A day as an agent · Geneva → Dakar**
The agent simulator · CHF 200 through four channels, today's rates
🛠 WORKSHOP · 50 MIN

2.5 **End-of-morning quiz**
Eight questions · launched in class · counts toward participation
✅ QUIZ · 10 MIN

BLOCK 3 · 13:30–15:00 APIs and open banking · the console in test mode

2.6 **APIs and open banking**

Request and response · keys, tokens, idempotency, webhooks · PSD2, AISP and PISP

🎤 TALK · 20 MIN

2.7 **API workshop in test mode**

Eight missions in a payment console

🛠 WORKSHOP · 55 MIN

BLOCK 4 · 15:15–16:45 the three-way negotiation · your fintech's flow

2.8 **The three-way negotiation**

Merchant, provider, network · three confidential cards

🎤 TALK · 10 MIN

2.9 **The deal · your fintech's flow**

The deal calculator · the project's payment flow and revenue model

🛠 WORKSHOP · 65 MIN

← Course page

reset my Day 2 progress on this device

SECTION 2.1 · BLOCK 1

🎤 TALK · 20 MIN

Anatomy of a payment

Instructor's talk. If it does not take place, read this section on your own, then do the sorting exercise at the bottom of the page.

THE FOUR-CORNER MODEL

Five parties behind every card payment

Cardholder	Issuer	Scheme	Acquirer	Merchant
you, with the card	the bank that issued the card	Visa, Mastercard	the merchant's bank or PSP	the shop, the website

"Four corners": cardholder, issuer, acquirer, merchant. The scheme (card network) sits in the middle: it sets the rules, routes the messages and computes each party's position. Between the merchant and the acquirer there is often a **payment service provider** (Stripe, Adyen, Worldline…).

THE MECHANISM · FROM THE CARD TO THE MERCHANT'S ACCOUNT

Authorisation, clearing, settlement — eight steps

▸ A COFFEE PAID CHF 4.80 CONTACTLESS

1 · At the terminal. The card (or the phone) generates a single-use *cryptogram* from a secret key embedded in the chip: copying the number is not enough to clone a chip card. In Apple Pay or Google Pay, the real number is even replaced by a *token* specific to the phone.

EXAMPLE

Your phone touches the café's terminal. The real card number `4000 1234 5678 9010` does not leave it: the phone sends its token `4895 3700 0000 1102` and a cryptogram `8B2F61C4079E3A15`, computed from the amount, the date and a random number supplied by the terminal. Replayed tomorrow, that cryptogram would be refused.

2 · Authorisation request. The terminal sends the acquirer a message in **ISO 8583** format: amount (480 centimes), currency, card number or token, cryptogram, merchant identifier.

EXAMPLE

The message, field by field: `MTI 0100` (authorisation request) · `DE2 4895370000001102` (the token) · `DE3 000000` (a purchase) · `DE4 000000000480` (4.80) · `DE49 756` (Swiss franc) · `DE22 07` (contactless) · `DE42 CAFEDUCOIN0001` (the merchant) · `DE55` (the cryptogram).

3 · Routing. The acquirer forwards to the scheme (VisaNet, Banknet at Mastercard), which reads the first digits of the card — the **BIN**, 6 to 8 digits that identify the issuer — and routes the message to the right bank, anywhere in the world.

EXAMPLE

The network reads `489537`, the first six digits: they name the bank that issued the card. The token is first translated back to the real number by the network's token service, then the message goes to that bank. All of it takes a few tens of milliseconds.

4 · The issuer's decision. In a few milliseconds: valid card? correct cryptogram? sufficient balance or limit? And a **fraud-detection model** scores the transaction (place, time, amount, customer habits). Coded response: `00` approved, `51` insufficient funds, `05` declined.

EXAMPLE

At the issuer: card valid until 08/28 ✓ · cryptogram verified ✓ · balance CHF 1,250.00 ≥ 4.80 ✓ · fraud score 12 out of 999 (usual: café, neighbourhood, 8:15) ✓. Response `00`, authorisation code `A7K3Q9`. The same payment at 3 a.m. in Lagos, with the same card an hour after Geneva, would have scored high and got a `05`.

5 · Back to the terminal in one to two seconds: "Payment approved". No money has moved yet — the issuer has only *reserved* the amount on the cardholder's account.

EXAMPLE

The terminal receives `MTI 0110`, `DE39 00`: "Payment approved" appears. In your banking app: "Café du Coin · −4.80 · *pending*". Your available balance drops; your booked balance does not, yet.

6 · Clearing (in the evening). The merchant or its provider sends the batch of the day's transactions. The scheme computes, for each bank, what it owes and what it is owed — interchange and scheme fees included.

EXAMPLE

At 11 p.m. the café sends its batch: 312 payments, CHF 1,486.40. For your coffee the network records: amount 4.80 · interchange 0.44% = 0.02 owed to the issuer · scheme fee 0.15% = 0.01. The centimes of millions of payments become one net position per bank.

7 · Settlement (D+1). Net positions are settled between settlement banks: the issuer pays the acquirer the amount of the purchases *minus the interchange*.

EXAMPLE

The next day your bank pays: 4.80 – 0.02 interchange = 4.78 for this coffee, folded into one net transfer to the café's bank together with those of all its other customers. On your statement the line goes from "pending" to "booked".

8 · The merchant is credited by the acquirer, minus the MDR, one to three days later. For several weeks, the cardholder can still dispute: that is the *chargeback* window.

EXAMPLE

Two days later the café receives 4.80 – 1.5% commission = **CHF 4.73**. Of the 0.07 withheld: 0.02 for your bank, 0.01 for the network, 0.04 for the acquirer, which also pays for processing. If you disputed this coffee during the chargeback window (often 120 days), the café could have to give back the 4.80.

✓ All 8 steps are shown

WHO EARNS WHAT

The merchant discount rate, broken down

Merchant discount rate (MDR) = interchange + scheme fees + acquirer's margin

On a CHF 100 purchase with a 1.5% MDR, the merchant receives CHF 98.50. The **interchange** (set by the scheme, paid by the acquirer to the issuer) pays the issuer for the risk and credit it carries; the **scheme fees** pay Visa or Mastercard; the rest is the **acquirer's margin**, which also pays for the technical processing.

- **In the EU**, Regulation 2015/751 caps interchange on consumer cards at **0.2%** (debit) and **0.3%** (credit).
- **In Switzerland**, there is no law: agreements with the Competition Commission (COMCO) brought domestic credit-card interchange down to **0.44%** on average (since 2017).
- Interchange is what funds neobanks' "free" cards: they are issuers, and earn the interchange on every payment.

SECURITY AND FRAUD

Strong customer authentication, 3-D Secure, and who pays for fraud

The European **PSD2** directive requires **strong customer authentication** (SCA) for most online payments: two factors out of *knowledge* (a code), *possession* (the phone) and *inherence* (fingerprint, face). For cards, it goes through **3-D Secure**: the issuer asks for confirmation in its app. Version 2 is *risk-based*: the issuer receives around a hundred data points (device, history, amount) and lets low-risk payments through without friction.

The liability shift: when a payment has been authenticated with 3-D Secure, fraud is borne by the issuer. Without authentication, the cardholder disputes, the issuer raises a *chargeback* and the merchant loses the sale.

WITHOUT A CARD

Instant payment, TWINT, QR-bill

- **Instant payment.** In Switzerland, the SIC system (operated by SIX for the Swiss National Bank) has processed instant payments since August 2024: credited within seconds, 24 hours a day, 7 days a week, as ISO 20022 messages. In the EU, Regulation 2024/886 requires banks to receive and then to send SEPA instant transfers (ten seconds at most) at the price of an ordinary transfer, with verification of the payee's name.
- **TWINT.** The app is linked to the user's bank account; payment is made by scanning a QR code (or showing one), authorisation is given in real time, and the merchant is credited by its acquirer. No card network in the middle: TWINT is its own scheme.
- **The QR-bill.** Since 30 September 2022, it has replaced the orange and red payment slips. The QR code contains no money: it contains a *pre-filled payment order* — IBAN, amount, reference — which the payer's banking app reads and submits after confirmation.

CHECK · 3 MIN

Who does what in a card payment?

› CARDHOLDER · ISSUER · SCHEME · ACQUIRER · MERCHANT

Receives the interchange

Cardholder | Issuer | Scheme | Acquirer | Merchant

Pays the merchant discount rate (MDR)

Cardholder | Issuer | Scheme | Acquirer | Merchant

Routes the request to the right bank using the BIN

Cardholder | Issuer | Scheme | Acquirer | Merchant

Approves or declines the payment

Cardholder | Issuer | Scheme | Acquirer | Merchant

Credits the merchant, one to three days later

Cardholder | Issuer | Scheme | Acquirer | Merchant

Bears the fraud on a payment authenticated with 3-D Secure

Cardholder | Issuer | Scheme | Acquirer | Merchant

Sets the interchange schedule (under the authorities' oversight)

Cardholder | Issuer | Scheme | Acquirer | Merchant

0 of 7 sorted

SECTION 2.2 · BLOCK 1

WORKSHOP · 55 MIN

Interchange simulator · IBAN and QR-bill

A · 25 MIN · WHO WINS, WHO LOSES

Vary the MDR, the interchange, the fraud and the volume

An online merchant in Geneva. Each slider changes how the money is split between the four corners. Answer the three questions below the table, then explain what you observed.

Average basket (CHF) 50

Transactions per month 2,000

Merchant discount rate 1.50%

Interchange 0.44%

Scheme fees 0.15%

Acquirer's processing cost 0.25%

Fraud rate 0.10%

Without 3-D Secure | With 3-D Secure

PER MONTH · VOLUME CHF 100,000	CHF	PER CHF 100
Merchant — MDR + fraud (chargebacks)	-1,600.00	-1.60
Issuer — interchange	440.00	0.44
Scheme — fees	150.00	0.15
Acquirer — MDR – interchange – scheme fees – processing	660.00	0.66
Fraud lost, borne by the merchant	**100.00**	**0.10**

With the starting settings (MDR 1.5%, interchange 0.44%, scheme 0.15%, processing 0.25%), how much does the acquirer keep on a **CHF 100** purchase?

… CHF Check

Still the starting settings, without 3-D Secure: 2,000 transactions of CHF 50 per month and 0.10% fraud. Total payment cost for the merchant, **per month**?

… CHF Check

Interchange falls to **0.20%** (the EU debit cap) but the merchant discount rate stays at 1.5%. How much does the acquirer keep on CHF 100?

… CHF Check

✍️ **What you observed: which slider moves the most money, and towards whom? What does 3-D Secure change, and for whom?**

e.g. Lowering interchange to 0.2% increases the acquirer's margin if the MDR does not fall; with 3-D Secure, fraud shifts from the merchant to the issuer…

B · 20 MIN · THE IBAN AND THE QR-BILL

The check that prevents sending money to the wrong account

A Swiss IBAN has 21 characters: **CH**, two **check digits**, five **institution** digits (the bank), twelve **account** digits. To check it (ISO 13616 standard): move the first four characters to the end, replace each letter with a number (A = 10 … Z = 35; C = 12, H = 17), and divide the resulting large number by 97. **An IBAN is valid if the remainder is 1.** To *compute* the check digits, do the same with "00" in their place, and take 98 minus the remainder.

▸ IBAN CHECKER, STEP BY STEP

CH93 0076 2011 6238 5295 7 Check

1. Move **CH93** to the end: `007620116238529 57CH93`
2. Letters become numbers: `007620116238529 57121793`
3. Remainder of division by 97: **1** → **valid IBAN**

Compute the two check digits of the Swiss IBAN whose remaining part is `0076 2011 6238 5295 7`: CH**??** 0076 2011 6238 5295 7. *(Tip: check "CH00 0076 2011 6238 5295 7" in the tool, then 98 – remainder.)*

… Check

A customer swapped the last two digits: `CH93 0076 2011 6238 5297 5`. What remainder do you get?

… Check

The **QR reference** (27 digits) ends with a check digit computed by “recursive modulo 10”: start with a carry of 0; for each digit, the new carry is entry (carry + digit) mod 10 of the table `0 9 4 6 8 2 7 1 3 5`; at the end, the check digit is (10 – carry) mod 10.

› QR REFERENCE CALCULATOR

`210000000003139471430000901` Compute

Successive carries: `0+2→4 · 4+1→2 · 2+0→4 · 4+0→8 · 8+0→3 · 3+0→6 · 6+0→7 · 7+0→1 · 1+0→9 · 9+0→5 · 5+0→2 · 2+3→2 · 2+1→6 · 6+3→5 · 5+9→8 · 8+4→4 · 4+7→9 · 9+1→0 · 0+4→8 · 8+3→9 · 9+0→5 · 5+0→2 · 2+0→4 · 4+9→6 · 6+0→7 · 7+1→3`

Check digit: **7** → full reference `210000000003139471430009017`

An SME numbers its invoices and derives the reference `000000000000000202610230001` (26 digits) from them. What is the 27th, the check digit?

… Check

Here is the content of a QR-bill, as the code carries it (example from SIX’s implementation guidelines). One line per field; empty lines are optional fields left blank.

```
SPC   ← type
0200   ← version
1   ← coding
CH4431999123000889012   ← creditor IBAN
S   ← address type
Robert Schneider AG   ← name
Rue du Lac   ← street
1268   ← number
2501   ← postcode
Biel   ← town
CH   ← country
·   ← (ultimate creditor, 7 lines)
·
·
·
·
·
·
1949.75   ← amount
```

```
20  CHF   ← currency
21  S   ← debtor: address type
22  Pia-Maria Rutschmann-Schnyder   ← debtor: name
23  Grosse Marktgasse
24  28
25  9400
26  Rorschach
27  CH
28  QRR   ← reference type
29  210000000003139471430009017   ← reference
30  Order of 15 June 2020   ← message
31  EPD   ← end of data
```

TRUE OR FALSE?

Line 4: this is a **QR-IBAN**, because its institution number (31999) lies between 30000 and 31999.

True False

With a QR-IBAN, you can use a free reference in the international format (SCOR) instead of the QR reference.

True False

The last digit of the reference (line 29) is a recursive modulo 10 check digit.

True False

Scanning this code with your banking app sends the money immediately, without confirmation.

True False

The amount line can be left empty: the payer then enters the amount themselves.

True False

The QR-bill replaced the orange and red payment slips, which have no longer been accepted since 30 September 2022.

True False

0 of 6 sorted

DEBRIEF · 10 MIN

Two questions: why does interchange exist, and who should pay it? Why does a TWINT payment or a QR-bill cost the merchant less than a card payment?

SECTION 2.3 · BLOCK 2

🎤 TALK · 15 MIN

Mobile money and transfers

Instructor's talk, or self-study reading.

THE MECHANISM · ELECTRONIC MONEY

Where does the money in a Wave or Orange Money wallet come from?

- **One-for-one issuance.** When a customer hands 10,000 FCFA in cash to an agent, the **electronic money issuer** records 10,000 FCFA in their wallet. In return, the issuer holds 10,000 FCFA in a **safeguarding account** at a bank. The sum of the wallets always equals this account: electronic money is not created, it is *converted*.
- **The wallet is a claim** of the customer on the issuer, not a bank deposit: the issuer cannot lend this money. That is the BCEAO rule in the WAEMU and the rule of Directive 2009/110/EC in the EU.
- **The agent** is a neighbourhood shopkeeper who keeps two stocks: **cash** in the drawer and **electronic money** (the *float*) on the phone. A *deposit* (cash-in) means receiving cash and giving up float; a *withdrawal* (cash-out), the reverse. The agent earns a commission per transaction.
- **Rebalancing.** When the float or the cash runs out, the agent must go to the bank or to a super-agent to convert one into the other — a trip, time, a risk of theft. An agent who runs short turns customers away: that is the first breaking point of a network.

The measured effect. In Kenya, access to M-Pesa (launched in 2007) lifted about 2% of households — some 194,000 — out of extreme poverty, with a stronger effect for female-headed households (Suri and Jack, *Science*, 2016).

THE MECHANISM · AN INTERNATIONAL TRANSFER

What a transfer really costs: the fees *and* the rate

Total cost = amount sent − (amount received ÷ mid-market rate)

The **mid-market rate** is the midpoint between buy and sell on the foreign-exchange market. A provider advertising "zero fees" often makes its money on the rate. Between the

CFA franc and the euro, the rate is **fixed**: €1 = 655.957 FCFA; so CHF 1 = 655.957 ÷ (price of one euro in francs) FCFA.

- **The bank (SWIFT transfer)**: the money goes through a chain of correspondent banks, each of which may deduct fees; one to three business days.
- **Wise**: the money does not cross the border. Wise receives your francs in its Swiss account and pays the recipient from its local account; flows in both directions offset each other. Mid-market rate, displayed fees.
- **To a mobile wallet**: a transfer service credits the recipient's Wave or Orange Money wallet directly; the recipient cashes out at an agent or pays with it.
- **A stablecoin**: francs → USDC on an exchange (fees and spread), transfer on the blockchain (network fees), USDC → FCFA with a local intermediary (spread). Each conversion has its cost (Day 4).

United Nations Sustainable Development Goal 10.c: bring the cost of remittances below **3%** by 2030. The global average measured by the World Bank is about **6%** for a 200-dollar remittance, and sub-Saharan Africa is the most expensive region.

CHECK · 3 MIN

What happens to the agent's two stocks?

▸ CASH ↑ FLOAT ↓ · CASH ↓ FLOAT ↑ · NOTHING CHANGES

A customer deposits 10,000 FCFA to send to her mother

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

A customer withdraws 25,000 FCFA received from his brother in France

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

Two customers send each other money from phone to phone

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

The agent goes to the bank to deposit surplus cash in exchange for float

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

A customer pays the shopkeeper next door by scanning his QR code

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

End of the month: a teacher withdraws the salary the State paid into his wallet

Cash ↑ float ↓ | Cash ↓ float ↑ | Nothing changes

0 of 6 sorted

SECTION 2.4 · BLOCK 2

🛠 WORKSHOP · 50 MIN

A day as an agent · CHF 200 from Geneva to Dakar

A · 20 MIN · THE AGENT NETWORK SIMULATOR

You run a Wave point in Dakar. Sixteen customers today.

You start with **150,000 FCFA** in cash and **150,000 FCFA** of float. Each transaction served earns you **1%**. Before each customer, you can **rebalance** (go to the super-agent and split half-and-half): **500 FCFA** in transport. A customer you cannot serve goes to the competitor: **–2,000 FCFA** (the commissions they will no longer bring you). The maximum possible is 4,950 FCFA.

CASH	FLOAT (ELECTRONIC MONEY)	TODAY'S EARNINGS
150,000	150,000	0
		0 CUSTOMER(S) LOST

Customer 1 of 16 · **Awa** wants to **deposit 20,000 FCFA**.

Serve the customer | **Rebalance first (–500)**

The day's log appears here.

✍ **Your score, and your strategy in one sentence: when did you rebalance, and why at that moment?**

The score is inserted at the end of the game; add your strategy.

B · 30 MIN · WITH TODAY'S REAL RATES

Sending CHF 200 from Geneva to Dakar through four channels

In pairs. For each channel, go to the provider's website or simulator *today*, simulate sending CHF 200 to Senegal, and note the fees, the exchange rate offered and the

announced delivery time. First enter today's euro rate in francs (from the SNB, Wise or your bank's website): the table derives the CHF → FCFA mid-market rate from it.

Price of one euro in francs today → **mid-market rate: CHF 1 = 705.33 FCFA**

0.93

CHANNEL	FEES (CHF)	RATE OFFERED (FCFA PER CHF 1)	DELIVERY TIME	RECEIVED (FCFA)	TOTAL COST (CHF)	COST %
Bank · international transfer (SWIFT)						
Wise						
To a Wave wallet						
Stablecoin (CHF → USDC → FCFA)						

Copy the table into my notebook ↓ | Fill with example values

The example values are fictitious; they only show how the table calculates. The task is to collect the real ones.

✍️ **The table, then your conclusion: which channel for which customer (who has an account, a smartphone, needs cash the same day)?**

SECTION 2.5 · BLOCK 2

✅ QUIZ · 10 MIN

End-of-morning quiz

INTERACTIVE QUIZ · COUNTS TOWARD PARTICIPATION

Eight questions, one attempt each, all explained

Launched in class; this page unlocks by itself. On your own, without an AI assistant. A missed quiz scores zero, unless arranged in advance.

🔒 **The quiz starts together.** It is launched in class; this page unlocks by itself within seconds, without reloading.

0 / 8 answered

1. **In the four-corner model, interchange is paid…**

- by the merchant to the scheme
- by the acquirer to the issuer
- by the issuer to the acquirer
- by the cardholder to their bank

2. **The merchant discount rate (MDR) breaks down into…**

- cardholder fees + tax
- interchange only
- issuer’s margin + VAT
- interchange + scheme fees + acquirer’s margin

3. **PSD2 strong customer authentication requires…**

- two factors out of knowledge, possession and inherence
- a twelve-character password
- a handwritten signature
- physical presence at a branch

4. An instant payment (SIC in Switzerland, SEPA Instant in the EU) is credited…

- the next business day
- at the end of the day, in batches
- within seconds, 24 hours a day, 7 days a week
- after manual approval by the bank

5. The funds backing an electronic money issuer’s electronic money are…

- deposited one-for-one in a safeguarding account at a bank
- lent to SMEs to finance the network
- invested in shares
- non-existent: electronic money is created by the operator

6. For a mobile money agent, a customer’s withdrawal (cash-out)…

- increases their cash and decreases their float
- does not change their stocks
- decreases both their cash and their float
- decreases their cash and increases their float

7. **In a payment API, a CHF 49.90 purchase is sent with…**

- "amount": 49.90
- "amount": 4990
- "amount": "CHF 49.90"
- "amount": 49

8. **An app that triggers a transfer from your bank account, with your consent, to pay for an online purchase is, in PSD2 terms…**

- an AISP
- a card issuer
- a PISP
- an acquirer

SECTION 2.6 · BLOCK 3

🎤 TALK · 20 MIN

APIs and open banking

Instructor's talk, or self-study reading.

WHAT AN API IS

A request, a response — like a form filled in by a program

An **API** (application programming interface) lets one program call another. Payment APIs are almost all **REST**: the program sends an HTTP *request* to an address (the *endpoint*) and receives a *response* in JSON.

```
POST https://api.stripe.com/v1/payment_intents          ←   method + endpoint
Authorization: Bearer sk_test_…                         ←   the secret key
Idempotency-Key: order-2026-1043                        ←   duplicate guard
{ "amount": 4990, "currency": "chf" }                       ←   the body

200 OK
{ "id": "pi_3Q…", "status": "requires_payment_method", "amount": 4990 }
```

- **Methods**: `GET` reads, `POST` creates or triggers, `DELETE` deletes.
- **Status codes**: `200` success · `400` malformed request · `401` not authenticated · `402` payment declined · `404` not found · `5xx` server-side failure.
- **Amounts in the smallest unit**: CHF 49.90 is written `4990` (centimes) — except for currencies without decimals, such as the CFA franc: 10,000 FCFA is written `10000`.

WHAT PROTECTS

Keys, tokens, idempotency, webhooks

- **Keys.** The *secret key* (`sk_test_…` in test, `sk_live_…` in production) gives full rights over the account: it stays on the merchant's server, never in a web page or an app. The *publishable key* (`pk_…`) may be in the page; it only allows card data to be transmitted.
- **Tokens.** The card number goes directly from the browser to the provider, which returns a *token* (`pm_…`) to the merchant. The merchant never sees the card: it falls almost entirely outside the scope of the PCI DSS security standard.
- **Idempotency.** If the connection drops during a payment, the program resends the request. With the same `Idempotency-Key`, the provider returns the result of the first

one instead of charging twice.

- **Webhooks.** Some results arrive later (3-D Secure confirmation, bank transfer). Instead of asking again and again, the merchant provides an address; the provider calls it when the status changes (`payment_intent.succeeded`). The call is signed, so that an attacker cannot announce a fake payment.

OPEN BANKING

PSD2: banks must open accounts to authorised third parties

Since PSD2 (Directive 2015/2366, applied since 2018), a European bank must give access to payment accounts, *with the customer's consent*, to two types of licensed providers:

- **AISP** — account *information* service provider: it *reads* balances and transactions (budget aggregator, accounting tool, credit scoring).
- **PISP** — payment *initiation* service provider: it *triggers* a transfer from the customer's account (paying online by bank transfer, without a card).

The customer gives consent in their bank's app (OAuth 2.0 flow, with strong customer authentication); the third party receives an *access token* limited in time and in rights, never the password. The APIs follow standards such as the Berlin Group's NextGenPSD2. **Switzerland has no legal obligation**: opening up happens through market agreements, notably via SIX's bLink platform. In the EU, the revision (PSD3 and the Payment Services Regulation) is in the process of adoption.

CHECK · 3 MIN

AISP, PISP, or neither?

▸ CLASSIFY EACH SERVICE

A budgeting app that shows your accounts at three banks together

AISP | PISP | Neither

A "pay by bank transfer" button on a merchant website that debits your account after confirmation in your bank's app

AISP PISP Neither

An online lender that analyses your last twelve months of transactions to decide on a loan

AISP PISP Neither

A neobank that issues you a debit card

AISP PISP Neither

Accounting software that executes an SME's supplier payments from its bank account

AISP PISP Neither

Paying in a shop with a Visa card

AISP PISP Neither

0 of 6 sorted

SECTION 2.7 · BLOCK 3

🛠 WORKSHOP · 55 MIN

API workshop · a payment platform in test mode

IN TEAMS · EIGHT MISSIONS

Create a customer, collect CHF 49.90, get declined, pass 3-D Secure, refund

The console below imitates Stripe's API in test mode, simplified: JSON body, fictitious ids, no real money, nothing leaves your browser. Choose an endpoint, click "Template" to pre-fill the body, edit it, send, read the response and the webhook log. Confirm and retrieve act on the last payment created (or on the one whose `"id"` you put in the body). The missions tick themselves off.

1. Create a customer (`POST /v1/customers`)
2. Create a payment of **CHF 49.90** for this customer — mind the unit
3. Confirm this payment with `pm_card_visa` and receive the success webhook
4. Create another payment and get it **declined** for insufficient funds — read the `decline_code`
5. Create a payment that requires **3-D Secure**, then have it confirmed
6. Send the same payment creation twice with the same `Idempotency-Key` — only one payment must exist
7. Trigger a **401** error by replacing the secret key with a publishable key `pk_test_…`
8. Refund the successful payment (`POST /v1/refunds`)

REQUEST

`POST /v1/customers`

Authorization: `Bearer sk_test_mfb545_demo`

Idempotency-Key: `(optional)`

```
{
  "name": "Awa Diop",
  "email": "awa.diop@example.com"
}
```

Template | Send →

Test payment methods: `pm_card_visa` (accepted) · `pm_card_chargeDeclinedInsufficientFunds` (declined, insufficient funds) · `pm_card_authenticationRequired` (3-D Secure required). These are the ones from Stripe's documentation; with a real Stripe account in test mode (free), the same requests work.

RESPONSE

```
_
```

WEBHOOKS RECEIVED BY YOUR SERVER

none yet

✍️ **Two short answers: why does the merchant wait for the webhook before shipping the order? What could someone do who finds your secret key in your website's code?**

e.g. The final status may arrive after 3-D Secure…; with the secret key one can issue refunds, read the customers…

DEBRIEF · 15 MIN

Which mission caused trouble, and which error message put you on the right track? Where, in your own fintech's customer journey, will a third-party API do part of the work?

SECTION 2.8 · BLOCK 4

🎤 INSTRUCTIONS · 10 MIN

Role play · the three-way negotiation

THE SITUATION

An online merchant, its payment provider and a card network

Teranga Market, a Geneva online shop (groceries and cosmetics for the West African diaspora), collects **CHF 1.2 million** a year by card. Its provider, **PayLéman**, charges it 1.8%. A competitor offers 1.4%. Online fraud reaches 0.4% of sales, above the monitoring threshold of the card network **Kora**. The three parties meet to set: **the MDR**, **who pays for fraud**, and **whether 3-D Secure is activated**.

The rules. The three project teams form three tables; at each table, one person per role (three per table). Read *only your own card*. 20 minutes of negotiation; a deal is recorded in the calculator in the next section. No deal is a possible outcome — to be compared with your fallback option.

🛒 The merchant

Teranga Market

🏦 The provider

PayLéman SA, acquirer

🌐 The network

Kora, card network

SECTION 2.9 · BLOCK 4

🛠 NEGOTIATION AND PROJECT · 65 MIN

The deal, then your fintech's payment flow

20 MIN · NEGOTIATION · 15 MIN · DEBRIEFING

Enter the terms of your deal: the calculator gives each party's result

⏱ Negotiation **19:56** Pause ↺

Agreed MDR 1.80%

Share of remaining fraud paid by the merchant 100%

3-D Secure activated

No

Network rebate (0.05%) if 3-D Secure

No

3-D Secure integration paid by

the merchant

Copy the terms into my notebook ↓

PER YEAR	RESULT	OBJECTIVE / FALLBACK	
🛒 Merchant — total cost (MDR + fraud + lost sales)	CHF 26,400	objective < 21,000 · fallback 24,600	✗ worse than the fallback
🏦 Provider — profit	CHF 10,200	objective ≥ 4,800 · fallback 0	✓ objective
🌐 Network — net fees	CHF 1,200	objective: fraud < 0.3%	fraud 0.40%

Without 3-D Secure: fraud 0.40% (CHF 4,800), network penalty of CHF 1,200 for the provider. As soon as you change a term, the deal is recorded.

✍ **Your role, the terms of the deal (or the lack of one), and the argument that moved the table.**

The calculator's button inserts the terms; add your role and the decisive argument.

At the debriefing, the three tables compare their deals. The question to ask: *was there a deal better for all three*? What did one need to know that only one role knew?

30 MIN · IN TEAMS · THE PROJECT

The payment flow and revenue model of your pension-savings app

Both teams' project: a pension-savings app for Senegal, Congo or a neighbouring country, with small monthly payments invested in Swiss blue-chip equity funds until 65. First draw the flow on paper — from the customer's mobile wallet to the fund units in Switzerland, then the way back at 65: who pays whom, by what means, how long it takes, who holds the money in the meantime — then describe it here. Each member fills in their notebook — the same version for the whole team.

1. The flow: who pays whom, on which rail (card, instant payment, TWINT, mobile money, stablecoin…), how fast, who holds the money in the meantime, and under which licence.

e.g. The customer pays 10,000 FCFA from Orange Money → collection account at a partner bank in Dakar (D+0) → monthly pooling, FCFA → CHF conversion → custodian bank in Switzerland → fund units bought in each customer's name (D+…); at 65: units sold → CHF → FCFA → mobile wallet…

2. The revenue model: who pays, the mechanism (a % of assets, a fee per payment, FX margin, subscription…), the price, and an estimated ARPU with your calculation. For an app that must earn trust, also say how each fee is shown to the customer.

e.g. 0.8% a year of assets, no hidden FX margin; average balance 300,000 FCFA → ARPU 200 FCFA a month; variable cost: …

BEFORE DAY 3 · SATURDAY 24 OCTOBER (THE NEXT DAY)

- Check that your Google account opens Google Colab ↗: tomorrow's Block 1 is a guided credit-scoring notebook.
- Day 3 adds the pension-savings plan itself: note how much your customer could pay in each month, from what age, and what would make them trust you — or run.